

Cover Lane Progress Summary

Route-1 L1 SAA-level singleton cover

Worktree: C:/Users/Chengsong/Documents/posix-route1/cover

Date: 2026-06-21

Target

The COVER lane target remains the Isabelle lemma:

```
lemma strong_apder_acc_RALTS_singleton_cover:
  "strong_apder_acc (RALTS rs) k
    <= (UN q:set rs. strong_apder_acc (RALTS [q]) k)"
```

The proof strategy follows ROUTE_COVER.md: work directly at the `strong_apder_acc` carrier level, split the carrier into the term part and the root part, and route cross-prune star escapes to the full singleton SAA carrier rather than to a root-only singleton source.

Current Status

- Latest private-heap Isabelle build is **GREEN**.
- Session: `Posix_Card_Route1_Cover`.
- Theory file: `r1cover/Card_Route1_Cover.thy`.
- Worktree is clean after the latest commits.
- No sorry, oops, or admit was introduced.

Private Build Command Used

```
& 'C:\Users\Chengsong\Isabelle2025-2\contrib\cygwin\bin\bash.exe' -lc
"export USER_HOME=/cygdrive/c/Users/Chengsong/Documents/posix-route1/cover/.isa_home
&& export HOME=$USER_HOME
&& cd /cygdrive/c/Users/Chengsong/Documents/posix-route1/cover
&& '/cygdrive/c/Users/Chengsong/Isabelle2025-2/bin/isabelle'
  build -d . Posix_Card_Route1_Cover"
```

Latest observed result:

Finished `Posix_Card_Route1_Cover`

Green Commits Added

Commit	Contribution
a5ce844	Added singleton escape tail credits: a direct RONE tail-credit lemma and a guarded RSTAR tail-credit lemma requiring <code>rsimpStrong_raw (RSTAR s) = RSTAR s</code> .
4492c47	Added <code>rsimp_ALTs</code> escape witnesses: <code>rsimp_ALTs_RONE_member</code> and <code>rsimp_ALTs_RSTAR_member</code> . These connect collapse cases back to actual head branches.
ee3931d	Proved <code>singleton_saa_ok</code> preservation through <code>tagged_rdistinct</code> and <code>tagged_rflts</code> , with the NF side condition needed for flattening ALT rows.
d062867	Added an NF-safe member-to-RALTS opening lemma and <code>singleton_saa_ok_flatten</code> .
caf0028	Added tagged strong-ALT bookkeeping and the opening bridge <code>row_dlforms_rsimpStrong_raw_RALTS_tagged_open</code> .
b4c84ea	Added the carrier split helpers: the term carrier cover, the RZERO/RONE root cases, and singleton-root closure helpers.

Green Lemma Inventory

The file currently contains the following green building blocks relevant to the target:

Area	Green lemmas
Term carrier	<code>strong_apder_acc_RALTS_terms_singleton_cover</code>
Unit root cases	<code>strong_apder_acc_RALTS_root_RZERO_singleton_cover</code>
	<code>strong_apder_acc_RALTS_root_RONE_singleton_cover</code>
	<code>strong_apder_acc_RALTS_singleton_cover_RZERO</code>
	<code>strong_apder_acc_RALTS_singleton_cover_RONE</code>
	<code>strong_apder_acc_RALTS_singleton_cover_RONE</code>
Tagged final rows	<code>tagged_Strong_ALTs_rows_origin</code>
	<code>tagged_Strong_ALTs_rows_snd_rtail_nf</code>
	<code>row_dlforms_rsimpStrong_raw_RALTS_tagged_open</code>
singleton_saa_ok transport	<code>singleton_saa_ok_flatten</code>
	<code>tagged_rdistinct_preserves_singleton_saa_ok</code>
	<code>tagged_rflts_preserves_singleton_saa_ok</code>
Escape witnesses and credits	<code>rsimp_ALTs_RONE_member</code>
	<code>rsimp_ALTs_RSTAR_member</code>
	<code>row_dlforms_RONE_tail_credit</code>
	<code>row_dlforms_RSTAR_tail_credit_fix</code>

Important Correction

The unguarded helper

```
row_dlforms (rsimp7_SEQ_atom (RSTAR s) (rsimpStrong_raw k))
```

```
<= row_dlforms (rsimpStrong_raw (rsimp4_SEQ_atom (RSTAR s) k))
```

is false without a source/fixpoint condition. A minimal shape is:

```
tail = RSTAR RZERO
k     = RONE
```

The file therefore keeps only the guarded green version:

```
assumes "rsimpStrong_raw (RSTAR s) = RSTAR s"
shows
  row_dlforms (rsimp7_SEQ_atom (RSTAR s) (rsimpStrong_raw k))
    <= row_dlforms (rsimpStrong_raw (rsimp4_SEQ_atom (RSTAR s) k))
```

This matches the intended B2 proof shape: the star escape must carry a real source/fixpoint witness from the strong/pruned row pipeline.

Open Proof Obligation

The target lemma itself is still **OPEN**. The remaining hard step is the nonunit root carrier:

```
row_dlforms (rsimpStrong_raw (rsimp4_SEQ_atom (RALTS rs) k))
  <= (UN q:set rs. strong_apder_acc (RALTS [q]) k)
```

after excluding $k = RZERO$ and $k = RONE$.

The next local lemma should combine the tagged-row opening with the precise prune escape characterization:

- B1: surviving root-origin rows are routed by `singleton_saa_ok`.
- B2: collapsed s^* escape rows are routed through the ACC side using `singleton_saa_key_credit`, the `rsimp_ALTs` witnesses, and the guarded `RSTAR` tail-credit lemma.

Next Step

Prove the precise prune-pair preservation lemma:

```
tagged_prune_pair_raw preserves singleton_saa_ok
  modulo a key-credit escape case
```

The proof should avoid the previously rejected root-only route:

```
dl_le_pruned_altseq
singleton_source_ok
```

It should also avoid broad `auto` over all regex constructors; the attempted wide association lemma timed out and needs constructor-specific Isar splitting.